

Semester: **Spring 2023**
Course Code: **CSE460**
Course Title: **VLSI Design**

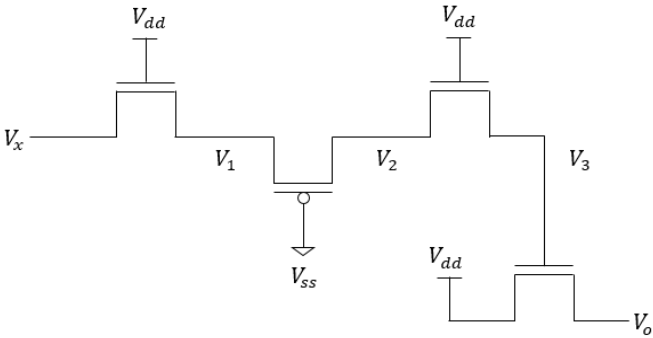
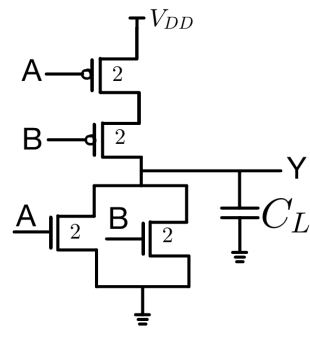
Final Exam
Full Marks: **15 x 3 = 45**
Time: **1 hour 40 minutes**
Date: **2nd May 2023**

Set A

Student ID:	Name:	Section:
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[Answer any **THREE** questions out of **FOUR**. Each question carries equal marks.]

[After the exam, the question paper should be turned in along with the answer script.]

1. (CO3)	 Figure 1	
	 Figure 2	
	(a) The pass transistor circuit in figure 1 has the following specifications: $V_{DD} = 5V, V_{ss} = 0V, V_{tp} = -0.5V, V_{tn} = 0.5V, V_o = 2.5V$ and $0.5V < V_x < 4.5V$	
	(i)	Calculate the unknown voltage values: V_x, V_1, V_2, V_3 [5]
	(b) A digital system-on-chip has the above gate from figure 2 implemented on it. Individual transistor widths $k_{nMOS} = 2$ and $k_{pMOS} = 2$ were chosen to achieve the effective rise and fall resistance equal to that of a unit inverter, R for the given gate. The system has $V_{DD}=1V$, activity factor, $\alpha=0.2$, unit transistor's width, $W = 6 \mu m$, load capacitor, $C_L = 2 pF$ and parasitic diffusion capacitance for both PMOS and NMOS transistors are $0.5 fF/\mu m$. Here, $1G = 10^9, 1p = 10^{-12}, 1f = 10^{-15}$.	
	(i)	Calculate the total capacitance at node Y. [4]
	(ii)	Estimate the Switching power when operating in 1.4 GHz. [3]
	(iii)	Determine if the system can switch <i>faster</i> or <i>slower</i>, when V_{DD} is <i>halved</i> while the switching power and the capacitance remain <i>same</i> [as (ii)]. [3]

2. (CO3) Consider the following circuit:

The circuit has the following specifications:

- **nMOS specifications:** $W_n = 1\ \mu m$, $L_n = 0.18\ \mu m$, $\mu_n C_{ox} = 50\ \mu A/V^2$, $V_{tn} = 0.4V$
- **pMOS specifications:** $W_p = 2\ \mu m$, $L_p = 0.18\ \mu m$, $\mu_p C_{ox} = 100\ \mu A/V^2$, $V_{tp} = -0.4V$
- **Other specifications:** $I_2 = 1mA$, $I_3 = 0.5mA$, $R_2 = 0.5k\Omega$, $R_3 = 2.5k\Omega$, $V_{dd} = 3V$, $V_{ss} = 0V$

(a)	Calculate the current constants (β_p, β_n) for both of the transistors.	[3]
(b)	Find the voltage across the R_3 resistance.	[3]
(c)	Determine the operating mode of the nMOS transistor.	[5]
(d)	Identify the value of R_1 in order to keep the pMOS transistor in the “linear” operating mode.	[4]

3. CO3) Observe the figure and the truth table and answer the following questions. Assume the effective rise and fall resistances of all the components are equal to that of a unit inverter (**R**) in the worst case.

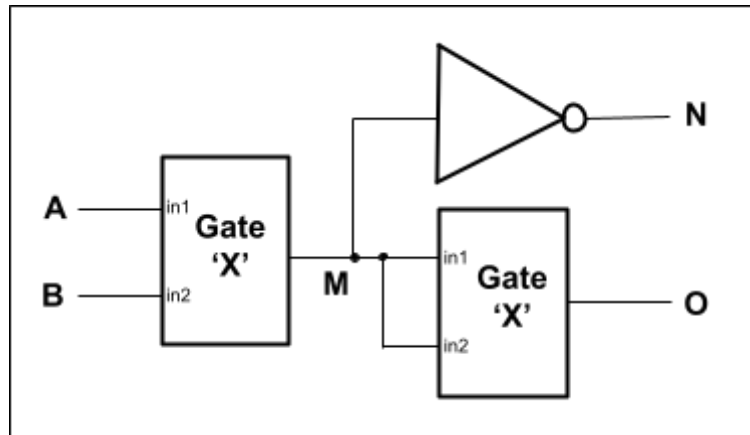
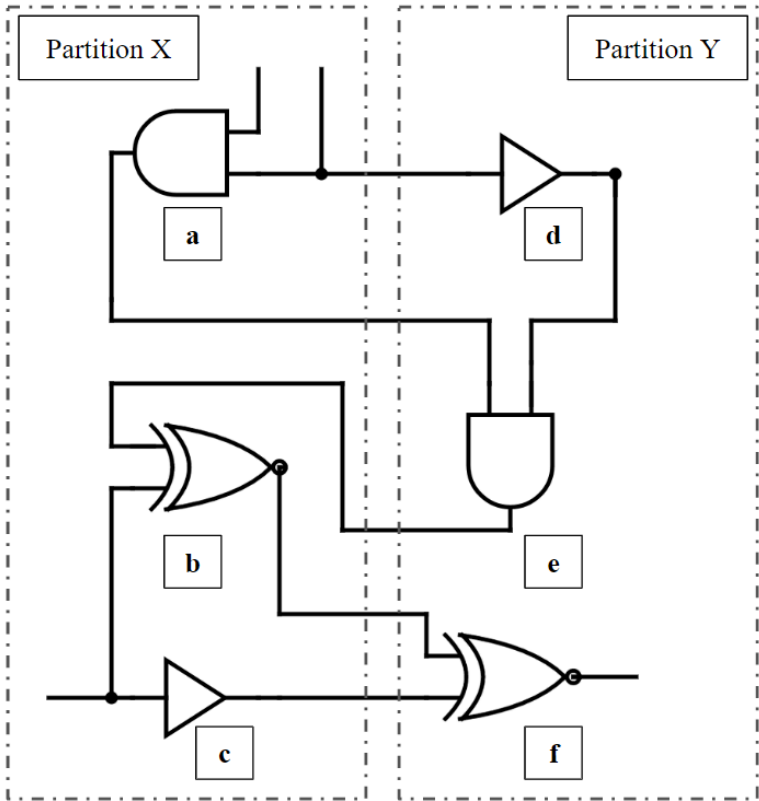


Figure: Two-input CMOS gate 'X' is driving another 'X' gate and an inverter.

Truth Table for node N

A	B	N
0	0	0
0	1	0
1	0	0
1	1	1

(a)	Identify gate 'X'.	[2]
(b)	Calculate the capacitances of nodes A, B, and M with the necessary figures.	[6]
(c)	Calculate 'propagation delay rise (t_{pdr})' at node M.	[4]
(d)	'N and O nodes have identical outputs but the N node is faster' - do you agree with the statement? Explain qualitatively.	[3]

4. (CO4, CO5)	Consider the following circuit with 6 logic gates (a-f) and the initial partitioning marked by the dotted lines.	
		
(a)	Represent the given circuit and the initial partitioning using a graph.	[3]
(b)	Compute the initial cut cost.	[1]
(c)	Perform the first pass of the algorithm. For every “i”th iteration inside the first pass, until all the nodes are swapped and fixed, do the following: - Compute the node costs of all unfixed nodes - Find the maximum gain of swapping a pair of nodes (Δg_i) - Swap the pair and draw the updated graph	[6]
(d)	Finish the first pass by computing the maximum positive gain, G_m . Suggest how many swaps should be actually executed in the first pass.	[4]
(e)	Should you perform subsequent passes of the algorithm? Why or why not?	[1]